NISHTHA SHELLY

Postdoctoral Researcher — 2D Materials & Photonic Devices
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Institut Quantique (IQ), Université de Sherbrooke, Québec, Canada
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PROFILE

Physicist with a PhD in photonic 2D-material devices and 4+ years of hands-on experience building and operating high-precision optical and electro-optic metrology systems — including custom differential reflectance and photo-current setups — across a broad cryogenic range (1.8 K–400 K). Experienced in handling ultra-high vacuum chambers, laser systems, optical tables, Python-based lab automation and data acquisition, SMU and lock-in-based electrical characterization and linking material-level metrology to device performance. Looking to bring this precision-measurement background and more to quantum metrology and calibration for next-generation quantum optics hardware.

WORK EXPERIENCE

Postdoctoral Fellow

Université de Sherbrooke | July 2025 – At present | Sherbrooke, Canada

Supervisor: Mathieu Massicotte

- Optimization of CVD-grown 2D materials transfer for reproducible, large-area, scalable adaptation for integrated photonic devices.
- Photonic study of magnons in 2D magnetic material tunnel junctions.
- TMD-based avalanche photodetectors.
- Supervising students (past and current) — interns: 3, master's: 2, PhD: 1.

Nano Innovator Ambassador

ATLANT 3D | At present | Copenhagen, Denmark (remote)

- Bridging the gap between potential partners/collaborators and ATLANT 3D's technology and products.

Doctoral Researcher

Trinity College Dublin, University of Dublin | Sept 2020 – Feb 2025 | Dublin, Ireland

Supervisors: David McCloskey, Christopher Murray

- Utilized a low-pressure CVD-like technique (Thermally-Assisted Conversion / CVT process) to fabricate large-area, continuous, transfer-free ultrathin and few-layer MoS_2/SiN/Au pixelated stacks.
- Demonstrated active refractive index tuning of few-layer MoS_2 in 0th and 1st order Fabry-Pérot interference stacks ranging from 5 µm to 200 µm laterally using a custom-built CCD-based differential reflectance (CCD-DR) metrology setup for mapping material optical properties.
- Experimentally demonstrated uniform differential reflectance across TCO/dielectric interfaces procured from collaborators at Tyndall National Institute, Cork, Ireland, using the CCD-DR setup.
- Utilized Hall bar and Van der Pauw designs to characterize commercially procured and in-house fabricated TCOs via precision electrical measurement.
- Implemented compatibility of the TAC process with standard photolithographic processes, proving uniformity and control over TAC-based TMD synthesis in the area of interest.
- Designed a custom-made mask for few-layer MoS_2-based TFTs using CleWin.

Student Research Assistant

Technical University of Denmark | Jul 2019 – Aug 2019 | Roskilde, Denmark

Supervisors: Nini Pryds, Yang Li

- Assisted in device testing and experimental data analysis of strain effect and post-deposition annealing on resistive random-access memory (RRAM) devices via profilometry and Imina microprobe station.

Graduate Researcher (M.Sc. Thesis)

Technical University of Denmark | Feb 2020 – July 2020 | Lyngby, Denmark

Supervisor: Nini Pryds

- Performed heat transfer modelling using COMSOL to understand heat distribution at the sample surface and its effect on 2DEG formation.
- Developed an experimental setup to enhance low-temperature electron mobility of complex oxide heterostructures via annealing and gating.
- Investigated a model explaining 2DEG-defect dynamics at the interface of the heterostructure.

Undergraduate Researcher (B.Tech Thesis)

Manipal Institute of Technology | Jan 2018 – May 2018 | Karnataka, India

Supervisor: Ashok Rao

- Studied the effect of sample preparation procedure and doping/compositing to link material properties with electrical properties.
- Prepared bulk samples of Cu_2SnSe_3 and $Bi_2Sb_{2-x}Te_3$ (x = 0.4, 0.45, 0.5) using the solid-state reaction method.
- Observed enhancement in electrical conductivity with greater number of sintering cycles, longer grinding duration, and doping.

EDUCATION

PhD, School of Physics (photonics)

Trinity College Dublin, University of Dublin | Sept 2020 – Feb 2025 | Dublin, Ireland

Thesis: *Electro-optic properties and applications of thermally-assisted conversion based few-layer MoS_2 devices*

M.Sc., Sustainable Energy — Energy Conversion and Storage

Technical University of Denmark | Sept 2018 – July 2020 | Lyngby, Denmark

Thesis: *Experimental platform for controlled migration of oxygen vacancies in $SrTiO_3$ heterostructures under elevated temperatures*

B.Tech, Electrical and Electronics Engineering

Manipal Institute of Technology | Aug 2014 – May 2018 | Karnataka, India

Thesis: *Synthesis of doped/composite thermoelectric materials for Peltier cooler applications*

SKILLS

Laboratory: Material synthesis (large-area CVD graphene transfer, vapour phase deposition — TAC/CVD, magnetron sputtering, material characterization (Raman, PL spectroscopy, XRR, AFM, ellipsometry, SEM, UV-Vis) | photodiode modelling, I-V and C-V characterization, cryogenic and differential reflectance measurement, photolithography (Microtech, Suss, OAI mask aligner, Heidelberg maskless direct writing), mask design (CleWin), wet/dry etching, class-3 lasers, optical workstations, wafer handling, cleanroom experience.

Software: AutoCAD, SketchUp, CASINO, FEM-COMSOL, CrystalMatch, FullProf Suite, ImageJ, LabVIEW, LaTeX, LTspice, MATLAB, OriginPro. First-principles calculations: DFT (GPAW).

Transferable: Setting up electro-optical experiments (CW and pulsed laser measurements); Python-based lab automation, data acquisition, instrument integration; 4+ years of Raman spectroscopy, AFM, spectroscopic ellipsometry and electro-optic characterization using custom-built CCD differential reflectance and frequency domain thermo-reflectance setups; photocurrent studies of avalanche photodiodes and magnetic materials; maintaining sample/calibration records; DOE and SOP documentation; collaborating with cleanroom managers, CTOs and technicians on recipes, tool monitoring and hardware/software debugging; lab tool manager training new users and mentoring students.

Equipment: Analog/digital lock-in amplifiers, trans-impedance amplifiers, SMUs, digital multimeters | Optical filters, polarizers, waveplates, dichroic mirrors, photo-elastic modulators, photodiodes; cryostat systems (1.8 K–400K), piezo stages, motors.

Languages: Fluent: English (CELPIP 2026: R-11, W-12, S-12, L-11), Hindi |Beginner: French, Danish.

PUBLICATIONS

- Invited to publish in the Special Issue of the journal Nanophotonics.
- N. Shelly, et al., "Strong Absorption and Modulation in Deeply Subwavelength Transition Metal Dichalcogenide Thin Films," (ongoing).
- N. Shelly, et al., "Enhanced Photoconductivity in Ultrathin MoS_2 Films Using Lithography Free Structural Colour," (ongoing).
- N. Shelly, et al., "Field Effect Modulation using Transparent Conductive Oxides and Multilayer Transition Metal Dichalcogenides," in CLEO 2023, Technical Digest Series (Optica Publishing Group, 2023), paper JW2A.97.
- E. Roy, N. Shelly, C. Murray, and D. McCloskey, "Investigation of Field Effect Tuning of Refractive Index in Transparent Conducting Oxide Thin Film Fabry-Perot Cavity Structures," in Proceedings of The 13th International Conference on Metamaterials, Photonic Crystals and Plasmonics, ISSN: 2429-1390, 2023.
- Soman, S., Shelly, N., Kurian, C. P., & Sudheer Kumar, T. S., "DC transformer modeling and control of DC-DC buck converter," International Journal of Power Electronics and Drive Systems, 10(1), 319-329, 2019.
- D. Khicher, E. Roy, N. Shelly, C. Murray and D. McCloskey, "Enhanced Coloration Efficiency and Cycling Stability in Ultrathin NiOx Films Utilizing Zeroth Order Fabry-Perot Resonance" — under review (ACS Applied Materials & Interfaces).

CONFERENCE TALKS & PRESENTATIONS

- "Optimization of graphene transfer assisted by a nickel layer" – Double axes meeting: Nanomaterials and Biologically-Inspired Materials & Materials for New Advanced Technologies, RQMP, McGill University, March 2026.
- "Optical modulation in lithographically defined MoS_2 pixels" (talk delivered by David McCloskey) — Nanop Barcelona, 2023.
- "Electro-optic Modulation in Large Area Multilayer MoS_2 Pixel Arrays" — ItsHer Women in STEMM Summit, Dublin, 2024.